DiagnoMath

AI-Powered Classroom Cognitive Diagnostic Assessment Platform

Organization
Centre for Educational Technology (CET), IIT Bombay

Project Type
Teacher Mobile Application

Project Leads
Prof. Ramkumar Rajendran & Aditya

Project Overview

DiagnoMath is an AI-powered classroom assessment platform that combines traditional paper-based assessments with Cognitive Diagnostic Modelling (CDM) to provide teachers with actionable insights into student learning. The system is designed specifically for classroom environments where students may not have access to personal mobile devices, allowing teachers to continue using familiar assessment practices while benefiting from automated diagnostic analysis. During a classroom assessment, the teacher displays multiple-choice questions using a projector, smartboard or classroom display. Students record their answers on printed response sheets or structured answer booklets instead of using digital devices. After the assessment, the teacher scans each student's response sheet using the DiagnoMath mobile application. The application automatically recognizes student responses using Optical Mark Recognition (OMR), digitizes the responses, dynamically constructs the appropriate Q-matrix, and applies the **Generalized Non-Parametric Classification (GNPC)** cognitive diagnostic model. Rather than simply calculating marks, the platform identifies the mathematical concepts and cognitive attributes each learner has mastered and those requiring further instructional support. Teachers receive individualized learner reports together with classroom-level analytics that support evidence-based remediation and differentiated instruction.

The Educational Challenge

Many schools continue to rely on paper-based classroom assessments because students either do not possess personal smartphones or classroom policies discourage their use. Although teachers can efficiently administer assessments to entire classes, evaluating responses beyond simple scoring remains time-consuming and rarely provides insight into students' conceptual understanding. Teachers need practical tools that preserve existing classroom workflows while providing meaningful diagnostic information about learner misconceptions and cognitive mastery.

Current Practice

Teachers present questions using classroom projectors, smartboards or printed worksheets while students record answers on paper. Response sheets are manually checked and marks are entered into spreadsheets or gradebooks. Diagnostic analysis depends entirely on teacher expertise and is rarely performed systematically. Existing digital assessment systems typically assume one device per learner, making them unsuitable for many classrooms where paper-based assessments remain the norm.

Target Users

Primary Users

- Mathematics teachers
- School students participating in classroom assessments
- Assessment designers
- Teacher educators

Secondary Users

- School administrators
- Educational researchers
- Curriculum developers
- Learning support specialists

Deployment Context

- Teacher mobile application
- Paper-based classroom assessments
- Projector or smartboard-supported classrooms
- Schools where students do not require individual digital devices

Solution Vision

Design a teacher-centred mobile application that augments existing classroom assessment practices with AI-powered cognitive diagnosis. Rather than replacing paper-based assessments, the platform should seamlessly integrate into teachers' existing workflows by allowing them to display questions in class, scan student response sheets using a mobile device, automatically digitize responses through Optical Mark Recognition (OMR), and generate detailed cognitive diagnostic reports using the Generalized Non-Parametric Classification (GNPC) model. The system should minimise teacher workload while providing actionable evidence that supports targeted remediation, differentiated instruction and data-informed classroom teaching.

Minimum Viable Product

- Teacher authentication
- Assessment creation interface
- Question bank management
- Cognitive attribute selection
- Dynamic Q-matrix generation
- Printable student response sheets
- Mobile answer sheet scanning
- Optical Mark Recognition (OMR)
- Automatic response digitization
- GNPC diagnostic engine
- Individual learner mastery reports
- Teacher classroom dashboard

Future Enhancements

- AI-generated remedial worksheets
- Automatic grouping of learners by misconceptions
- Personalized intervention recommendations
- Longitudinal mastery tracking
- Curriculum coverage analytics
- School-wide performance dashboards
- Printable parent reports
- PDF and Excel export
- Offline scanning mode
- Learning Management System integration
- Multi-subject support
- Teacher recommendation engine

Expected Impact

- Preserve familiar paper-based classroom assessment practices while introducing AI-powered analysis.
- Eliminate manual data entry through automatic OMR-based response digitization.
- Provide teachers with concept-level mastery information instead of only marks.
- Enable rapid classroom diagnosis immediately after assessments.
- Support targeted remediation based on identified cognitive attributes.
- Reduce teacher workload associated with assessment analysis and reporting.
- Make Cognitive Diagnostic Modelling practical for everyday school classrooms.
- Improve evidence-based mathematics instruction through actionable learner analytics.

Design Challenge

- How might teachers perform sophisticated cognitive diagnosis without changing their existing classroom assessment practices?
- How can paper-based answer sheets be accurately converted into digital assessment data using mobile scanning?
- How should GNPC-based mastery reports be presented so they are understandable and immediately useful for teachers?
- How can diagnostic information be translated into classroom interventions rather than simply reporting learner performance?
- How might the platform operate effectively in classrooms where students do not have access to personal digital devices?
- How can teachers quickly identify common misconceptions across an entire class using cognitive diagnostic analytics?
- How should future AI-generated remediation build upon learner mastery profiles while remaining transparent and pedagogically meaningful?